FINAL PROJECT REPORT

Heart Disease Analysis

Heartbeat of Data — A Visual Journey Through Heart Disease Risk Factors

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Project Name	Heart Disease Analysis Using Tableau Visualization
Date	4 March 2026
Tools Used	Tableau Public, MySQL, Python, Flask
Dataset	Heart_new2.csv — 4,500 rows, 18 columns
GitHub	https://github.com/Souravkadam/Heart-Disease-Analysis

1. INTRODUCTION

1.1 Project Overview

The Heart Disease Analysis project is a data visualization initiative that leverages Tableau Public to explore and analyze heart disease risk factors across a dataset of 4,500 patient health records. The project transforms raw CSV health data into 10 interactive visualizations, one comprehensive dashboard, and a ten-point Tableau Story titled 'Heartbeat of Data'.

Heart disease remains one of the leading causes of mortality worldwide. According to the National Heart, Lung and Blood Institute, the most important risk factors include obesity, sedentary lifestyle, smoking, and diabetes. This project addresses the challenge of making complex multi-variable health data accessible and actionable through visual analytics.

The dataset — Heart_new2.csv — contains 18 clinical and lifestyle variables including BMI, smoking status, alcohol consumption, age category, gender, race, diabetic status, physical activity, general health rating, and indicators for stroke, asthma, kidney disease, and skin cancer. Out of 4,500 records, 483 patients have confirmed heart disease (10.7%) and 4,017 do not (89.3%).

1.2 Purpose

The purpose of this project is to:

- Provide cardiologists like Dr. Sharma with an interactive tool to identify high-risk patient groups based on lifestyle and demographic data.
- Enable health policy analysts like Ramesh to analyze heart disease prevalence across demographics and recommend evidence-based health policies.
- Allow individuals like Anita (45-year-old with family history) to compare personal health risk factors against population benchmarks.
- Demonstrate the power of Tableau Public as a business intelligence platform for healthcare data visualization without requiring advanced data science expertise.
- Publish a live, publicly accessible dashboard and story on Tableau Public for 24/7 stakeholder access.

2. IDEATION PHASE

2.1 Problem Statement

Cardiologists, health policymakers, and individuals lack a centralized, visual, and interactive tool to analyze heart disease risk factors from multi-variable patient data. Raw patient records across 18 variables — including BMI, smoking, age, gender, diabetes, and physical activity — are difficult to interpret without proper visualization, leading to delayed or uninformed clinical and policy decisions.

The three key problem statements identified through the ideation process:

PS No.	I am...	Trying to...	But...	Because...	Which makes me feel...
PS-1	Dr. Sharma (Cardiologist)	Identify highrisk patients based on lifestyle factors	Data is scattered and difficult to interpret	No centralized Tableau dashboard exists	Overwhelmed, unable to design prevention campaigns
PS-[1]	Ramesh (Health Analyst)	Analyze heart disease prevalence across demographics	Data is not visual or interactive	Spread across static reports without visualizations	Limited in drawing policy insights
PS-3	Anita (45yr Individual)	Monitor personal heart disease risk factors	Cannot compare metrics with healthy benchmarks	No simplified personalized health dashboard	Anxious and uninformed about risk reduction steps

[1] .2 Empathy Map Canvas

The empathy map was developed for the primary user persona — a healthcare professional and health-conscious individual seeking to understand heart disease risk factors.

Element	Details
WHO	Dr. Sharma (Cardiologist), Ramesh (Health Analyst), Anita (45-year-old individual with family history of heart disease)
GOAL	Understand, analyze, and prevent heart disease by identifying key risk factors through interactive Tableau dashboards built on 4,500 patient records
THINK & FEEL (Pains)	Difficulty identifying high-risk patients; scattered data across multiple sources; no visual tools; fear of missing early warning signs in patient data
THINK & FEEL (Gains)	Centralized dashboard; targeted prevention campaigns; actionable lifestyle insights; informed personal health decisions
HEAR	Colleagues discussing rising heart disease cases; media reports on lifestyle diseases; concerns about BMI and diabetes links to cardiovascular health
SEE	Increasing BMI and smoking patients; correlation charts in published research; Tableau dashboards used in other health domains
SAY & DO	Need better visual tools for patient data; want to compare metrics with healthy benchmarks; attend health awareness programs; monitor personal health metrics

2.3 Brainstorming

The following visualization ideas were generated and prioritized during the brainstorming phase:

Must-Do Visualizations (High Priority):

- Gender vs Heart Disease — stacked bar chart showing disease distribution by sex
- Age Category vs Heart Disease — grouped bar chart across 13 age groups
- BMI & Diabetic Analysis — treemap showing average BMI by age and diabetic status
- Smoking & Alcohol Impact — bar chart showing 4-combination lifestyle risk analysis

Should-Do Visualizations (Medium Priority):

- Race-wise Heart Disease Distribution — pie chart across 6 racial groups
- Physical Activity vs Heart Disease — donut chart (Active: 3,231 | Inactive: 1,269)
- General Health vs Heart Disease — bubble chart across 5 health rating categories
- Stroke co-occurrence — with Asthma, Kidney Disease, and Skin Cancer

Nice-to-Have (Future Scope):

- Real-time database connection via Flask API
- Predictive risk score calculator using machine learning

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey map captures the end-to-end experience of users interacting with the Heart Disease Analysis Tableau dashboard across five phases:

Phase	Experience	Interactions	Goals	Pain Points	Opportunity
Entice	Learns about HD risks from social media & blogs	Health ads, awareness campaigns, YouTube videos	Understand heart disease risk factors	Overwhelmed by medical jargon	Clean single dashboard entry point
Enter	Searches for risk factors, BMI ranges, smoking impact	Health websites, Tableau Public	Identify risk factors by age, gender, lifestyle	Difficult to compare multiple factors	Enable multi-factor filters
Engage	Opens dashboard, filters by gender/age/race	Tableau filters, bar charts, bubble charts	Make confident observations from 4,500 records	Treemap too complex for nontechnical users	Add tooltips, improve visual hierarchy
Exit	Narrows down key risk factors	Chart comparisons, story review	Final confirmation of findings	No predictive individual risk scoring	Add summary cards with top risk factors
Extend	Shares findings with peers and policymakers	Tableau Public link sharing, presentations	Contribute to health policy and awareness	Uncertain about data freshness	Add data refresh and dataset expansion options

3.2 Solution Requirements

Functional Requirements:

FR No.	Functional Requirement	Sub Requirement
FR-1	View Heart Disease Patient Data	Filter by age category, gender, race, and BMI range across 4,500 records
FR-2	Analyze Lifestyle Risk Factors	Smoking vs heart disease, alcohol impact, physical activity correlation

FR-3	View Disease Cooccurrence	Display stroke, diabetes, kidney disease, asthma alongside heart disease
FR-4	Dashboard & Story Interaction	KPIs, interactive filters, switch between different risk factor views
FR-5	Publish & Share	Publish to Tableau Public, share interactive story link with stakeholders

Non-Functional Requirements:

NFR No.	Requirement	Description
NFR-1	Usability	Dashboard is user-friendly for non-technical healthcare users
NFR-2	Security	Read-only data, published securely on Tableau Public
NFR-3	Reliability	Consistent data across all 10 visualizations, no load errors
NFR-4	Performance	Dashboard loads under 5 seconds on average internet
NFR-5	Availability	Hosted 24/7 via Tableau Public cloud
NFR-6	Scalability	New datasets or indicators can be added without complete redesign

3.3 Data Flow Diagram

The DFD Level 0 (Context Diagram) shows the Heart Disease Analysis system as a single process receiving Heart_new2.csv as input and delivering interactive dashboards to three stakeholders.

DFD Level 1 details six internal processes:

1. 1.0 Data Collection — User receives Heart_new2.csv (4,500 rows, 18 columns)
2. 2.0 Data Cleaning — CSV validated, whitespace removed, categorical values standardized
3. 3.0 Database Storage — Data stored in MySQL database (D1)
4. 4.0 Create Visualizations — 10 Tableau sheets created from MySQL data (saved to D2: Tableau Workbook)
5. 5.0 Publish Dashboard — Dashboard published to Tableau Public
6. 6.0 View & Interact — Stakeholders (Dr. Sharma, Ramesh, Anita) access live dashboard

3.4 Technology Stack

S.No	Component	Description	Technology Used
1	User Interface	Tableau Dashboard accessed via web browser	Tableau Public (Web UI)
2	Application Logic-1	Data filtering, sorting, and KPI calculations	Tableau Calculated Fields
3	Application Logic-2	Dashboard interactions — filters, stories, drill-downs	Tableau Sheets / Story Functions
4	Database	Stores heart disease patient records (4,500 rows, 18 cols)	MySQL
5	File Storage	Raw data files (.csv) — Heart_new2.csv	Local Filesystem
6	Web Framework	Flask web application serving the HTML frontend	Python Flask
7	Infrastructure	Tableau Public on Local PC + Tableau Public Cloud	Local PC, Tableau Cloud

4. PROJECT DESIGN

4.1 Problem Solution Fit

Element	Details

Target Customer	Cardiologists, Government Health Policy Analysts, and health-conscious individuals with risk factors
Observed Problem	Difficulty analyzing heart disease risk factors across 18 variables from 4,500 records without a centralized visual tool
Why It's a Problem	Healthcare professionals must manually browse raw CSV data causing confusion, time delays, and missed correlations
Your Solution	Comprehensive Tableau dashboard with 10 interactive visualizations, combined dashboard, and story titled Heartbeat of Data
Why It Works	Centralizes 18 risk factor variables using clear, interactive visuals and filters — accessible without data science expertise
Validation	10 visualizations clearly reveal: males 55-74 at highest risk, smokers show higher HD rates, diabetics have elevated stroke risk

4.2 Proposed Solution

Parameter	Description
Problem Statement	Lack of a centralized visual tool to analyze 18 heart disease risk variables across 4,500 patient records
Idea / Solution	Interactive Tableau dashboard with 10 visualizations and a Tableau Story for narrative-driven analytics
Novelty / Uniqueness	Integrates 18 clinical + lifestyle variables in one platform — demographic analysis + lifestyle risk + disease co-occurrence in a single Tableau Story
Social Impact	Empowers cardiologists to identify high-risk groups; enables policy analysts to plan interventions; helps individuals take proactive health steps
Business Model	Non-commercial academic project. Future potential: B2B subscriptions for hospital networks; custom analytics for government health departments
Scalability	New datasets from different regions or time periods can be added to MySQL; new visualizations added without redesigning existing architecture

4.3 Solution Architecture

The Heart Disease Analysis system follows a 5-layer architecture from raw data ingestion to stakeholder access:

Layer	Component	Description	Technology	Output
Layer 1	Data Source	Heart_new2.csv — 4,500 rows, 18 columns	Local Filesystem	Raw CSV Input
Layer 2	Data Processing	Data imported into MySQL; SQL used for cleaning and querying	MySQL	Structured Data
Layer 3	BI Visualization	Tableau connects to MySQL; 10 sheets created across chart types	Tableau Public	Visual Analytics
Layer 4	Dashboard & Story	All sheets assembled into Dashboard + Story with interactive filters	Tableau Story	Interactive Output
Layer 5	End Users	Published to Tableau Public; accessed via browser by all stakeholders	Tableau Cloud	Stakeholder Access

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog:

Sprint	Epic	Story No.	User Story	Points	Priority	Status
Sprint1	Data Collection & DB	USN-1	As a team, we can import Heart_new2.csv into MySQL, verify schema and all 18 columns	3	High	Done
Sprint1	Gender vs HD	USN-2	As a user, I can view heart disease count by gender in a stacked bar chart	2	High	Done
Sprint1	Age vs HD	USN-3	As a user, I can view HD distribution across 13 age categories by gender	2	High	Done
Sprint1	Smoking & Alcohol	USN-4	As a user, I can view how smoking & alcohol combinations affect heart disease	2	High	Done

Sprint2	BMI & Diabetic	USN-5	As a user, I can view treemap of age groups and avg BMI by diabetic status	2	Medium	Done
Sprint2	Race-wise HD	USN-6	As a user, I can view pie chart of HD prevalence across 6 racial groups	2	Medium	Done
Sprint2	Physical & Gen Health	USN-7	As a user, I can view physical activity donut (3231/1269) and general health bubble	2	Medium	Done
Sprint2	Stroke Cooccurrence	USN-8	As a user, I can view stroke vs diabetes scatter and stroke vs disease bar chart	2	High	Done
Sprint2	Dashboard & Story	USN-9	As a user, I can view all 10 sheets in combined Tableau Dashboard and Story	3	High	Done
Sprint2	Publish	USN-10	As a team, we can publish the .twbx workbook to Tableau Public	1	High	Done

Sprint Summary:

Sprint	Story Points	Duration	Start Date	End Date	Status
Sprint-1	14 Points	7 Days	4 March 2026	10 March 2026	Completed
Sprint-2	12 Points	7 Days	11 March 2026	17 March 2026	In Progress
TOTAL	26 Points	14 Days	—	—	—

Velocity: Sprint-1 completed 14 story points in 7 days = 2.0 story points/day. Sprint-2 estimated at 12 ÷ 2.0 = 6.0 days, planned for 7 days.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

S.No	Parameter	Screenshot / Values
1	Data Rendered	Total Records: 4,500 rows | Total Columns: 18 | Target: HeartDisease Yes:483, No:4,017 | CSV loaded directly into Tableau Public | All 18 columns rendered correctly as Dimensions and Measures
2	Data Preprocessing	Step 1: Collection (Heart_new2.csv) → Step 2: Data understanding (18 cols classified) → Step 3: Validation (no nulls/duplicates) → Step 4: Cleaning (standardized labels) → Step 5: CSV loaded into Tableau via Text File connection → Step 6: Transformation (COUNT, AVG, filters) → Step 7: Feature engineering (calculated fields, colour encoding)
3	Utilization of Filters	HeartDisease (Yes/No), Sex (Female/Male), AgeCategory (13 groups), Race (6 groups), Diabetic (4 values), Physical Activity (Yes/No), GenHealth (5 values) — all interactive filters connected across dashboard sheets
4	Calculation Fields Used	COUNT(HeartDisease), AVG(BMI) for treemap, CNT(HeartDisease)=4500 for donut, COUNT(Stroke) for scatter, SUM(PhysicalActivity), Colour encoding: Blue=No, Orange=Yes
5	Dashboard Design	No. of Visualizations: 4 | Gender vs HD (Stacked Bar), Age vs HD (Grouped Bar), Race-wise HD (Pie), Stroke vs Other Diseases (Bar) | Published: public.tableau.com/views/dashboard_17726415994650/Dashboard1
6	Story Design	No. of Visualizations: 10 | Story 1: Gender wise HD | Story 2: Age wise HD | Story 3: Diabetics vs Stroke | Story 4: Smoking & Alcohol | Story 5: Stroke vs Other Diseases | Story 6: Race-wise HD | Story 7: General Health | Story 8: Physical Activity | Story 9: Age & BMI vs Diabetic | Story 10: Triple Threat | Published: public.tableau.com/views/Book_17726265026180/Story1

7. RESULTS

7.1 Output Screenshots

The following 10 visualizations were successfully created and published on Tableau Public as part of the Heart Disease Analysis project:

#	Visualization Title	Chart Type	Key Finding
1	Gender vs Heart Disease	Stacked Bar Chart	Males: 256 Yes (13.5%) vs Females: 227 Yes (8.7%) — males at higher risk
2	Age vs Heart Disease	Grouped Bar Chart	Peak heart disease in 65-69 age group; risk increases with age from 40+
3	Diabetics vs Stroke	Scatter Plot	Diabetic patients (Yes) show significantly higher stroke counts than non-diabetics
4	Smoking & Alcohol vs HD	Horizontal Bar	Non-smokers dominate (2,390 No/No); smokers without alcohol: 139 HD cases
5	Stroke vs Other Diseases	Bar Chart	Stroke patients show co-occurrence with asthma, kidney disease, skin cancer
6	Race-wise HD Distribution	Pie Chart	White: 75.57%, Black: 17.81%, Other: 3.73%, Hispanic: 0.62%
7	General Health vs HD	Bubble Chart	Good health: 172 HD cases; Poor: 92; Fair: 140 — self-rating not sufficient predictor
8	Physical Activity vs HD	Donut Chart	Active: 3,231 (71.8%) vs Inactive: 1,269 (28.2%) out of 4,500 total patients
9	Age & BMI vs Diabetic	Treemap	BMI range 23.91-42.78; higher BMI clusters in older diabetic age groups
10	Triple Threat Analysis	Grouped Bar	Older age + diabetic + heart disease combination reveals highest risk profile

Dashboard URL: https://public.tableau.com/views/dashboard_17726415994650/Dashboard1

Story URL: https://public.tableau.com/views/Book_17726265026180/Story1

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Interactive and user-friendly — non-technical stakeholders can explore 18 risk factors without any coding knowledge.
- Publicly accessible 24/7 via Tableau Public — no installation required for end users.
- Combines 18 clinical and lifestyle variables in a single platform — rare in existing free healthcare dashboards.
- 10 diverse chart types (bar, pie, donut, treemap, bubble, scatter) provide multi-dimensional insights.
- Flask web integration allows embedding Tableau dashboards in a custom branded web application.
- Easily scalable — new patient datasets from different regions can be added to MySQL without redesign.
- Real-time filter interactions across all connected sheets in the dashboard.
- Published on Tableau Public cloud with 99.9% uptime guarantee.

Disadvantages:

- Tableau Public makes data publicly visible — not suitable for private patient data in clinical settings.
- No predictive analytics — the dashboard shows historical patterns but cannot predict individual risk scores.
- Dataset limited to 4,500 records — findings may not be fully generalizable to entire populations.
- CSV-based data connection requires manual refresh when new data is available — no real-time data streaming.
- Some complex visualizations like the treemap may be difficult to interpret for non-technical users.
- Tableau Public has a 15 GB storage limit and requires internet connection to view dashboards.

9. CONCLUSION

The Heart Disease Analysis project successfully demonstrates the power of Tableau Public as a business intelligence platform for transforming raw healthcare data into actionable visual insights. By building 10 interactive visualizations, one comprehensive dashboard, and a 10-point Tableau Story, the project delivers a complete analytical solution for three distinct stakeholder groups.

Key findings from the analysis reveal that males aged 55-74 show the highest heart disease rates, with diabetic patients demonstrating significantly elevated stroke risk. Physical inactivity affects 28.2% of patients, while poor general health self-ratings correlate with higher heart disease prevalence. The race-wise analysis shows White patients accounting for 75.57% of confirmed heart disease cases in the dataset.

The project meets all six functional requirements — from data filtering and lifestyle risk analysis to disease co-occurrence visualization and Tableau Public publication. All six non-functional requirements including usability, security, reliability, performance, availability, and scalability are addressed through the Tableau Public platform and MySQL backend architecture.

The Flask web application further extends the project by embedding both the dashboard and story within a professional branded website, making the analytics accessible to a broader audience. The complete solution is published on Tableau Public and hosted on GitHub, demonstrating a full end-toend data analytics project lifecycle from ideation through development to deployment.

10. FUTURE SCOPE

The Heart Disease Analysis project has significant potential for future enhancement and expansion:

- Machine Learning Integration: Implement a predictive risk score model using logistic regression or random forest to calculate individual heart disease probability based on the 18 input variables. Integrate the model with Flask for real-time risk prediction via a web form.
- Real-Time Data Streaming: Connect the Tableau dashboard to a live database with real-time patient data feeds from hospital management systems, enabling dynamic updates without manual CSV refreshes.
- Expanded Dataset: Incorporate regional health survey data from multiple countries to enable cross-population comparisons. Add temporal data to enable time-series analysis of heart disease trends across years.
- Natural Language Querying: Integrate Tableau's Ask Data feature or a custom NLP interface allowing stakeholders to ask questions in natural language and receive visual answers from the dashboard.
- Mobile Application: Develop a mobile-responsive version of the Flask application with a simplified personal risk calculator for individuals like Anita to assess their heart disease risk on the go.
- Electronic Health Record Integration: Connect with EHR systems like Epic or HL7 FHIR APIs to provide clinicians with patient-specific risk dashboards within their existing clinical workflow.
- Community Health Mapping: Add geographic visualization showing heart disease prevalence by region, state, or district to support targeted public health interventions.

11. APPENDIX

11.1 Source Code

Flask Application (app.py):

```
import pickle
from flask import Flask, render_template, request

app = Flask(__name__)

@app.route('/') def home():     return
render_template('index.html')

if __name__ == '__main__':
app.run(debug=False)
```

11.2 Dataset Link

Dataset: https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view File: Heart_new2.csv

- Records: 4,500 rows
- Features: 18 columns
- Target Variable: HeartDisease (Yes: 483 | No: 4,017)

11.3 GitHub & Project Demo Links

Resource	Link
GitHub Repository	https://github.com/Shripatil9082/Heart-Disease-Analysis
Tableau Dashboard	https://public.tableau.com/views/dashboard_17726415994650/Dashboard1?:language=enGB&:sid=
Tableau Story	https://public.tableau.com/views/Book_17726265026180/Story1?:language=enGB&:sid=&:redirect=

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